GLIDE Version 2.0: User Manual



https://github.com/YamKwongSang/GLIDE2.0

Guang-Sheng Ren
@CUGB

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1. Introduction and download

GLIDE is a linear inversion program for modeling exhumation rates from thermochronological data. The original GLIDE program was developed by Fox et al. (2014) in Fortran. Version 2.0 was subsequently rewritten in Python and optimized by Ren et al. (2026).

The basic modeling principles were described by Fox et al. (2014) and later by Willett et al. (2021). For details of the optimization, see Ren et al. (2026). In the following sections, the Shanxi Rifts system is used as an example, with thermochronological and geothermal data take from Wang et al. (2025).

GLIDE 2.0 is available at https://github.com/YamKwongSang/GLIDE2.0 and there are two ways to download the program:

1. Python code version: download via "Code" → "Download ZIP";
2. exe-file version: download via "Release" and choose the latest exe version (link of releases).

This manual is attached to the program (link of the manual).

2. Major process of modeling

2.1 Prepare files

A topography file (.xyz) and a thermochronological data table (.csv) are required for a GLIDE model. Make sure the two files are ready for a new model task.

An .xyz topography file is a regular grid of elevations and contains three columns of data, separated by spaces and without headers: longitudes (°), latitudes (°) and elevations (in meters). Positive longitude and latitude values represent east longitude and north latitude, respectively. For the program, the grid points must be ordered from west to east within each row, with the rows arranged from north to south. The grid in the .xyz file should be larger than the modeled region. For example, the following grid (also see data\Shanxi.xyz) is established for the topography between 108.00–117.00 °E and 33.00–41.50 °N:

```
108.000000   41.500000   1616
108.010000   41.500000   1588
108.020000   41.500000   1585
108.030000   41.500000   1561
******
116.990000   41.500000   1028
117.000000   41.500000   1121
108.000000   41.490000   1491
108.010000   41.490000   1460
******
116.990000   33.000000   22
117.000000   33.000000   22
```

This file can be generated from a local DEM file (raster) or download directly through the Google Earth

Engine. As for the later way, an extra function is attached to the main program and see section 3.1.

The thermochronological .csv table must include the following six columns: “longitude” (in degrees), “latitude” (in degrees), “elevation” (in m), “age” (in Ma), “std” (in Ma) and “method”, while additional columns are allowed. Among them, “age” and “std” are the thermochronological age and its one-sigma uncertainty, using Ma as the unit. The “method” column contains one of the following eight thermochronological methods:

“AFT” for apatite fission track;

“ZFT” for zircon fission track;

“AHe” for apatite (U-Th)/He;

“ZHe” for zircon (U-Th)/He;

“HAr” for hornblende Ar-Ar;

“MAr” for muscovite Ar-Ar;

“BAr” for biotite Ar-Ar;

“KAr” for K-feldspar Ar-Ar.

The above eight abbreviations are not case-sensitive while handling (e.g., Aft, AhE or Mar can be recognized). The parameters used to calculate closure temperatures are listed in Table 1, following Reiners and Brandon (2006) (E_a: activation energy; Ω: a proportionality constant derived from the diffusion parameters and effective spherical radius; symbol: corresponding method symbol in the original Fortran code):

Table 1. Thermochronological parameters used in GLIDE

method	E_a (kJ/mol)	Ω (s^{-1})	reference	symbol
AFT	147	2.05×10^{6}	Ketcham et al., 1999	1
ZFT	208	1.24×10^{8}	Brandon et al., 1998	2
AHe	138	7.64×10^{7}	Farley, 2000	3
ZHe	169	7.03×10^{5}	Reiners et al., 2004	4
HAr	268	1.32×10^{3}	Harrison, 1982	5
MAr	180	3.91	Hames and Bowring, 1994	6
BAr	197	733	Grove and Harrison, 1996	7
KAr	183	5.39×10^{5}	Foland, 1994	–

An example of a thermochronological data table is shown below (also see data\Shanxi.csv; provided by Wang et al. (2025)):

```
sample, latitude, longitude, elevation, age, std, method, reference
Thm208, 38.63795494, 112.4546794, 1646, 139, 10, AFT, Cao et al. (2015)
LLS12, 37.22705, 111.22745, 1276, 84.3, 7.7, ZFT, Li and Song (2010)
FP-10, 39.0926, 114.6728, 381, 18, 1.3, AHe, Chang et al. (2019)
FP-10, 39.0926, 114.6728, 381, 70.3, 5.7, ZHe, Chang et al. (2019)
******
```

The “sample”, “reference”, and other additional columns are optional and are provided for data

management purposes.

2.2 Open the program

Run the executable file GLIDE2.0.exe or the Python version GLIDE2.0.py to launch the GLIDE 2.0 program (might be slow for the first time). Running the .exe file does not require a Python environment. To run the Python version, Python ≥ 3.13 and *NumPy* (≥ 2.3) and *Numba* (≥ 0.62) packages are recommended. Users can install related packages through the file requirements.txt, using the following command:

```
pip install -r requirements.txt
```

The interface of the main program is shown in the Figure 1.

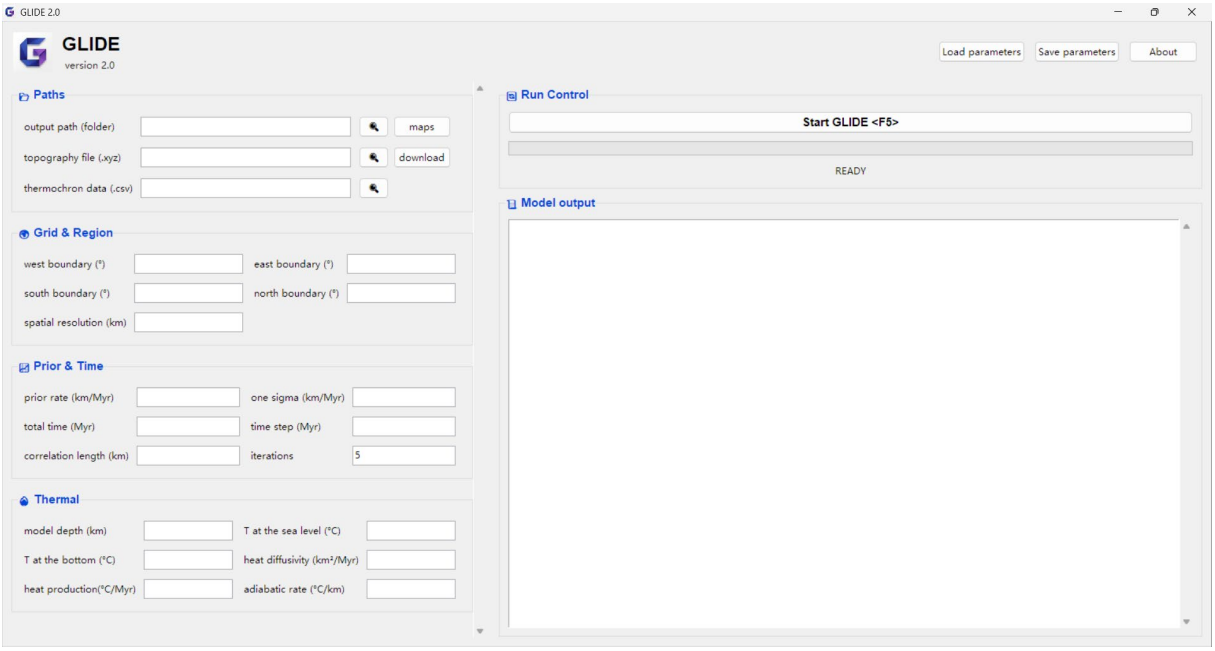


Figure 1. The interface of GLIDE main program, without inputs or outputs.

The left panel shows model-related paths and parameters and the right panel shows the run control and model outputs. The upper-right corner provides three options: loading parameters, saving parameters, and viewing information about the program.

2.3 Input parameters

The parameters and data used for a GLIDE modeling should be entered in the left panel, including the following four parts.

Paths:

Output path: a folder; for saving the output results by the program. See section 2.6 for details of output results.

Topography file: an .xyz file; records the topographic information. See section 2.1 for detailed requirements.

Thermochron data: a .csv table; records the thermochronology dataset. See section 2.1 for detailed requirements.

Grid & Region:

West boundary: in degrees (°); the westernmost longitude of the modeled region. Notice the four boundaries should be located inside of the .xyz topographic file.

East boundary: in degrees; the easternmost longitude of the modeled region.

North boundary: in degrees; the northernmost latitude of the modeled region.

South boundary: in degrees; the southernmost latitude of the modeled region.

Spatial resolution: in km; the expected spatial resolution of the model results, i.e., the approximate spacing of the control-point grid. The actual spacing may differ slightly from this value. This parameter does not affect the predicted ages. Therefore, it can be set to a coarser value to reduce computation time when evaluating the misfit and R^2, or to a finer value when detailed spatial information on exhumation is required. In the original Fortran code, its default setting is 3.0 (km).

Prior & Time:

Prior rate: in km/Myr (numerically equivalent to mm/yr); the prior exhumation rate of the modeled region, which needs to be estimated from the regional average before modeling. Regions with little thermochronological ages tend to obtain exhumation rates close to the prior rate.

One sigma: in km/Myr; the one-standard-deviation uncertainty of the prior exhumation rate, which should also be estimated from the regional data.

Total time: in Myr; the total time of modeling, i.e., the time when the model starts.

Time step: in Myr; the time step in modeling.

Correlation length: in km; the length used to evaluate the spatial correlations. To ensure the spatial continuity of the exhumation rate maps, it is recommended to be larger than the spatial resolution, and it can be set to 10 or 20 (km) in most studies. Larger correlation lengths tend to produce lower goodness-of-fit values.

Iterations: the number of the inversion iterations. The default is 5 and it is stable for most cases.

Thermal:

Model depth: in km; the depth of the model.

T at the sea level: in °C; the temperature at regional or adjacent sea level.

T at the bottom: in °C; the temperature of the model bottom, which determines the initial geothermal gradient (= (T at bottom – T at sea level) / model depth).

Heat diffusivity: in km^2/Myr; the heat diffusivity of the model. In most cases, it can be set to 25 or 30 (km^2/Myr).

Heat production: in °C/Myr; the heat production expressed as an equivalent temperature increase rate, mostly set to 0.

Adiabatic rate: in °C/km; the average regional adiabatic rate, due to adiabatic effects. It usually varies from 5.0 to 7.0 (°C/km) in the nature. In the original Fortran code, its default setting is 6.0 (°C/km).

After the parameters are entered, the interface will appear as shown in Figure 2., which is the same as the Shanxi Rifts case of Ren et al. (2026).

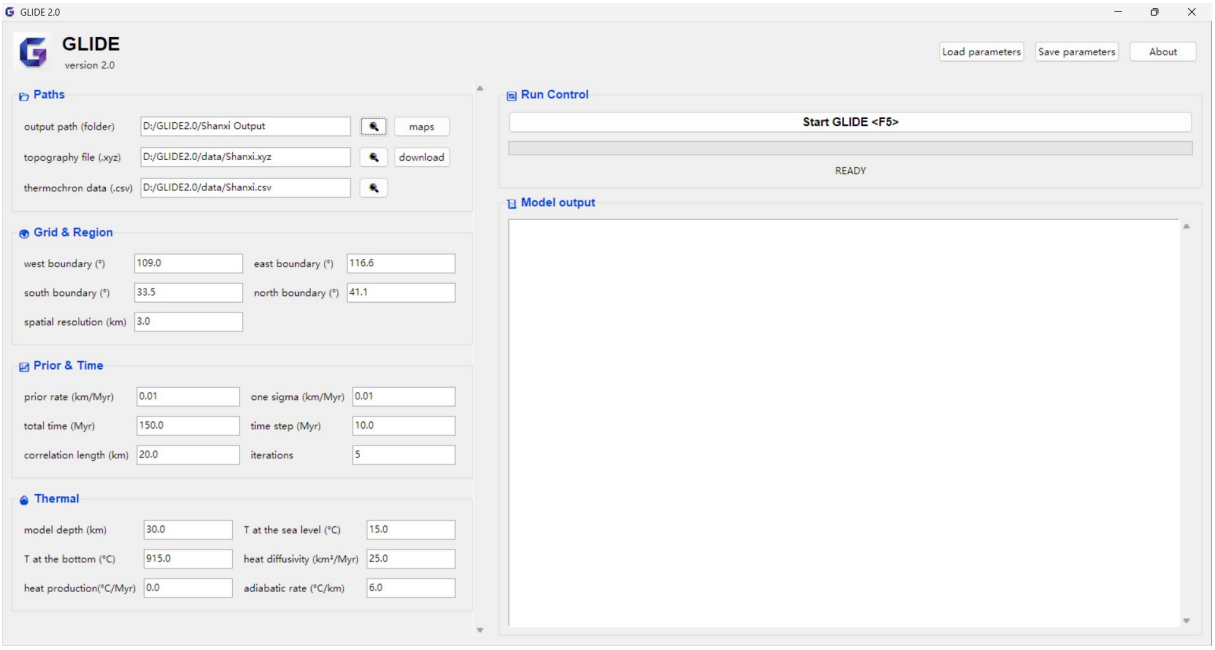


Figure 2. The interface after parameters are inputted.

2.4 Load/save parameters

For inputting and saving the parameters more easily, the “Load parameters” and “Save parameters” buttons in the upper-right corner can be used to load parameters from or save parameters to a .txt file, with the shortcut key of Ctrl + O and Ctrl + S accordingly. A saved parameter file is (see data\Shanxi param.txt):

```
# output path (folder)
D:/GLIDE2.0/Shanxi Output
# topography file (.xyz)
D:/GLIDE2.0/data/Shanxi.xyz
# thermochron data (.csv)
D:/GLIDE2.0/data/Shanxi.csv
# west / east / south / north boundary (°)
109.0 116.6 33.5 41.1
# spatial resolution expected for model (km)
3.0
******
```

The parameter file can be edited for a new model, if necessary, but the line order must not be changed. Meanwhile, the lines started with “#” have no effect on input (same as Python comment syntax).

2.5 Start modeling

After completing the preparation steps above, the model can be started. To run the model, click the “Start GLIDE” button or press shortcut key F5. And then, the progress bar will start and the state will change from “READY” to “RUNNING”.

Once finished, a small window will pop up and the state will change to “FINISHED” and relevant output information, including the prior misfits and R^2 and the posterior misfits and R^2 (Figure 3).

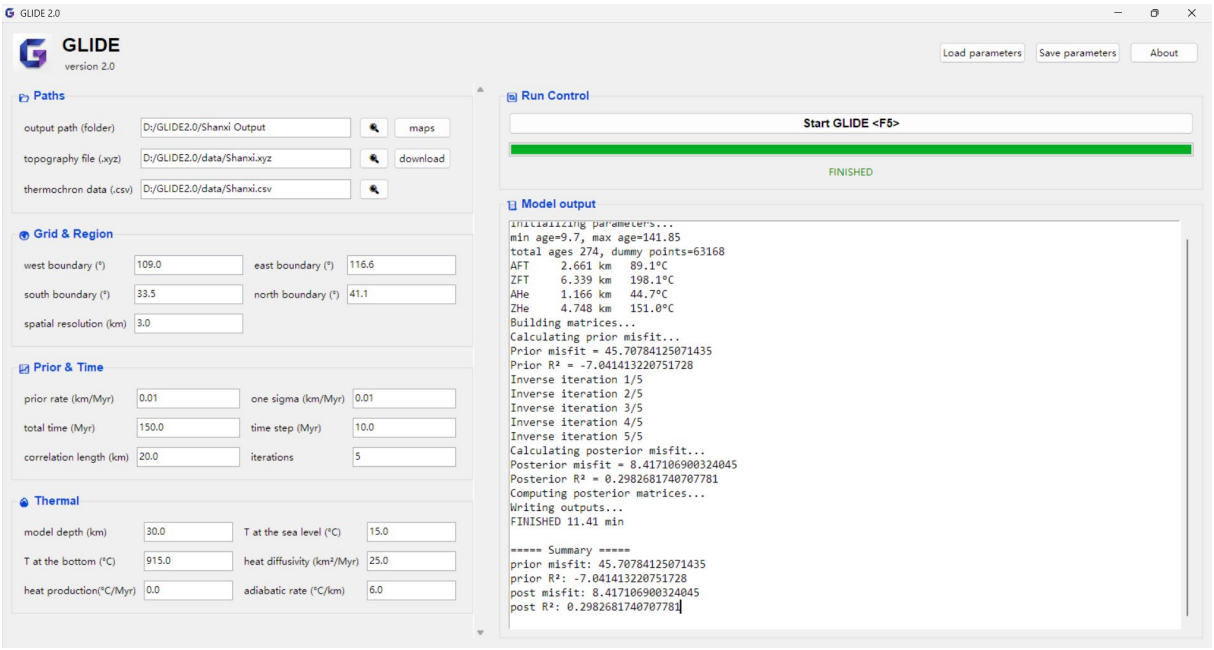


Figure 3. The GLIDE interface of after modeling is completed.

The output information is shown below. The misfit is defined as the weighted root-mean-square residual between observed and predicted thermochronological ages, and it can be influenced by both the number of age data and the model’s reliability. When the misfit is relatively large (e.g., > 10–15), the misfit itself becomes less informative, and R^2 may provide a more useful measure of model performance.

```
Initializing parameters...
min age=9.7, max age=141.85
total ages 274, dummy points=63168
AFT      2.661 km      89.1°C
ZFT      6.339 km      198.1°C
AHe      1.166 km      44.7°C
ZHe      4.748 km      151.0°C
Building matrices...
Calculating prior misfit...
Prior misfit = 45.70784125071435
Prior R² = -7.041413220751728
```

```
Inverse iteration 1/5
Inverse iteration 2/5
Inverse iteration 3/5
Inverse iteration 4/5
Inverse iteration 5/5
Calculating posterior misfit...
Posterior misfit = 8.4171069003240045
Posterior R² = 0.2982681740707781
Computing posterior matrices...
Writing outputs...
FINISHED 11.41 min
===== Summary =====
prior misfit: 45.707841250714 35
prior R²: -7.0414132207517 28
post misfit: 8.4171069003240045
post R²: 0.2982681740707781
```

2.6 Results

The output consists of a series of CSV tables containing the spatial distributions of posterior exhumation rate, reduced variance, and time resolution, together with a CSV table of predicted ages. Reduced variance ranges from 0 to 1 and reflects the relative uncertainty reduction achieved by the inversion. Time resolution represents the temporal resolving power of the inversion at each spatial node.

The output files are named after the end time (in Ma) of every stage. For example, the distribution of exhumation rate (in km/Myr), reduced variance and time resolution for the 10–0 Ma interval are given as (see Shanxi Output\0100.csv):

```
longitude, latitude, exhumation rate, reduced variance, time resolution
109.0,33.5,0.012700049495961188,0.9993823696200393,0.010702379104494186
109.0,33.52695034919901,0.01305541343998307,0.999210620128133,0.012106760489685899
109.0,33.553900698398024,0.013451690430919743,0.9989948860949841,0.01367173160088031
109.0,33.58085104759703,0.013892131655200389,0.9987254450891675,0.015409751155476017
******
```

And the predicted ages (in Ma) are given as (see Shanxi Output\Predicted ages.csv):

```
observed age, observed std, predicted age, method
139.0,10.0,117.78497895647415,AFT
60.0,5.0,58.36977631351052,AFT
82.0,6.0,79.34729733046484,AFT
68.0,6.0,69.56155695484094,AFT
```

******

3. Extra functions

There are two extra functions for downloading topography files (as .xyz) and mapping the modeled results. Neither function is required for GLIDE modeling. Both functions are optional and are provided for convenience only.

3.1 Get topography file

An independent Python script file, download.py, is provided along with the GLIDE main program (Figure 4). It uses the Google Earth Engine (GEE) API to allow users to download an .xyz file directly, without requiring local DEM data. To run it, the NumPy and earthengine-api packages must be installed, and a GEE account and project must be set up. The two packages can be installed using the following command:

```
pip install numpy earthengine-api
```

Additionally, the DEM data source can be selected from NASA, Copernicus and SRTM. If current Python environment meets the requirements, simply click the “download” button in the main program to execute it; otherwise, run download.py directly in user’s own environment.

This program is not mandatory for a GLIDE model. Some details of it may change because of GEE service. Users can also extract this .xyz file through a locally saved DEM file.

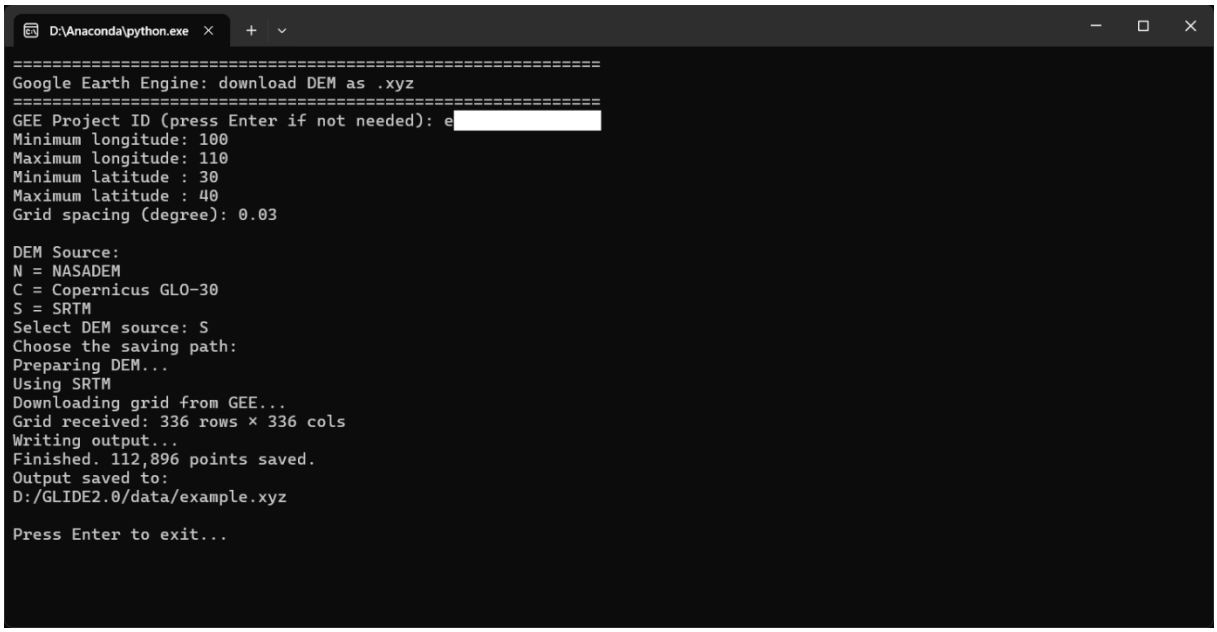


Figure 4. The interface of download.py, with inputs and outputs.

3.2 Map the results

An independent Python script file, mapper.py is also provided along with the GLIDE main program (Figure 5). NumPy and Matplotlib are required to run mapper.py. The two packages can be installed from a

command using:

```
pip install numpy matplotlib
```

If the current Python environment meets the requirements, simply click the “Maps” button in the main program to execute it; otherwise, run mapper.py directly in user’s own environment.

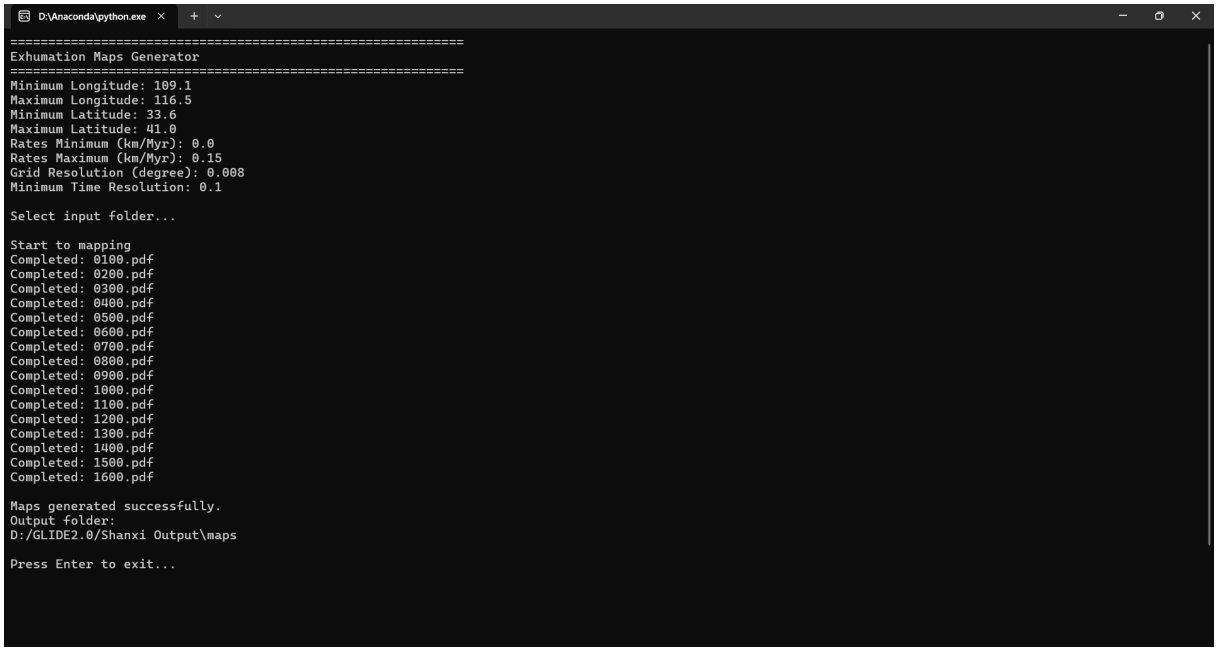


Figure 5. The interface of mapper.py, with inputs and outputs.

The following interface shows the inputs for mapper code and one of the maps (Figure 6, using the folder Shanxi Output\). The default resolution of the maps is 300 dpi.

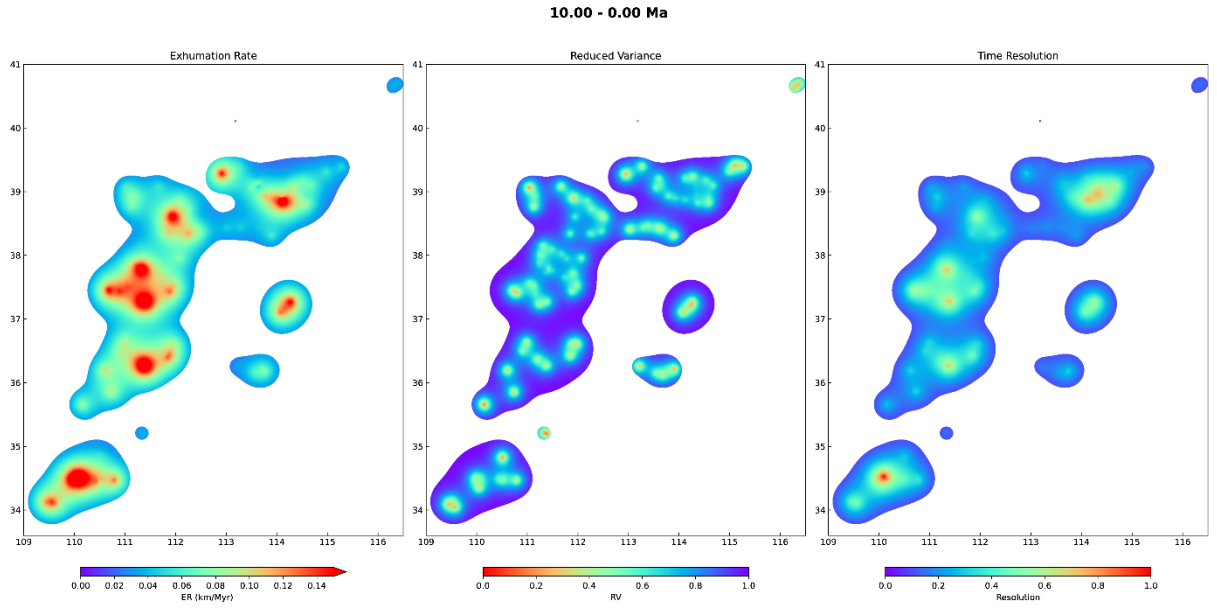


Figure 6. Generated maps for the 10–0 Ma interval of the Shanxi Rifts case.

References

Brandon, M.T., Roden-Tice, M.K., Garver, J.I., 1998. Late Cenozoic exhumation of the Cascadia accretionary wedge in the Olympic Mountains, northwest Washington State. Geological Society of America Bulletin 110, 985–1009. https://doi.org/10.1130/0016-7606(1998)110%3C0985:LCEOTC%3E2.3.CO;2

Farley, K.A., 2000. Helium diffusion from apatite: General behavior as illustrated by Durango fluorapatite. Journal of Geophysical Research 105, 2903–2914. https://doi.org/10.1029/1999JB900348

Foland, K.A., 1994. Argon diffusion in feldspars. In Feldspars and their reactions (pp. 415-447). Dordrecht: Springer Netherlands. https://doi.org/10.1007/978-94-011-1106-5_11

Fox, M., Herman, F., Willett, S.D., May, D.A., 2014. A linear inversion method to infer exhumation rates in space and time from thermochronometric data. Earth Surface Dynamics 2, 47–65. https://doi.org/10.5194/esurf-2-47-2014

Grove, M., Harrison, T.M., 1996. 40Ar* diffusion in Fe-rich biotite. American Mineralogist 81, 940-951. https://doi.org/10.2138/am-1996-7-816

Hames, W.E., Bowring, S.A., 1994. An empirical evaluation of the argon diffusion geometry in muscovite. Earth and Planetary Science Letters 124, 161-169. https://doi.org/10.1016/0012-821X(94)00079-4

Harrison, T.M., 1982. Diffusion of 40Ar in hornblende. Contributions to Mineralogy and Petrology 78, 324-331. https://doi.org/10.1007/BF00398927

Ketcham, R.A., Donelick, R.A., Carlson, W.D., 1999. Variability of apatite fission-track annealing kinetics; III, Extrapolation to geological time scales. American Mineralogist 84, 1235–1255. https://doi.org/10.2138/am-1999-0903 Reiners, P.W., Brandon, M.T., 2006. Using Thermochronology to Understand Orogenic Erosion. Annual Review of Earth and Planetary Sciences 34, 419–466. https://doi.org/10.1146/annurev.earth.34.031405.125202

Reiners, P.W., Spell, T.L., Nicolescu, S., Zanetti, K.A., 2004. Zircon (U-Th)/He thermochronometry: He diffusion and comparisons with 40Ar/39Ar dating. Geochimica et Cosmochimica Acta 68, 1857–1887. https://doi.org/10.1016/j.gca.2003.10.021

Ren, G.S., Han, X., Dai, J.G., Yu, Z.C., Fox, M., 2026. A Python-based optimization of the linear inversion method for thermochronological exhumation rates. Unpublished.

Wang, H., Li, S., Yang, X., Zuza, A.V., Liu, F., Sheldrick, T.C., He, X., Zhang, J., 2025. Exhumation history of the Shanxi Rift, Central Asia, derived from low-temperature thermochronology. Geological Society of America Bulletin 137, 1315–1334. https://doi.org/10.1130/B37693.1

Willett, S.D., Herman, F., Fox, M., Stalder, N., Ehlers, T.A., Jiao, R., Yang, R., 2021. Bias and error in modelling thermochronometric data: resolving a potential increase in Plio-Pleistocene erosion rate. Earth Surface Dynamics 9, 1153–1221. https://doi.org/10.5194/esurf-9-1153-2021